

Project Overview

Goal:

Move from a working prototype to a stable, scalable sensor platform and ultimately a public DIY kit.

Roadmap

Sprint	Objective	Output	Status
Sprint 2	Build stable platform and infrastructure	GitLab, MQTT, dashboard, PCB, working prototype	In Progress
Sprint 3–4	Deliver public DIY kit	Kit, enclosure, documentation	Planned

Sprint Overview:

HBO-ICT Studio schedule 2025-2026		
Semester 2		
Date	Lesson Week	Activity
02/Feb	1	Intake & Onboarding
09/Feb	2	Intake & Onboarding
16/Feb	3	Intake & Onboarding
23/Feb	4	Voorjaarsvakantie
02/Mar	5	Sprint 1
09/Mar	6	Sprint 1 (no educational activities)
16/Mar	7	Sprint 1
23/Mar	8	Sprint 2
30/Mar	9	Sprint 2
06/Apr	10	Sprint 2
13/Apr	11	Sprint 3
20/Apr	12	Sprint 3
27/Apr	-	Meivakantie
04/May	13	Sprint 3
11/May	14	Studio Next (no educational activities)
18/May	15	Sprint 4
25/May	16	Sprint 4
01/Jun	17	Sprint 4
08/Jun	18	High Stake / Repairs
15/Jun	19	Repairs
22/Jun	20	Repairs / High Stake Repairs
29/Jun	21	N/A

Sprint 2 — Development Environment

Objective

Build a stable, documented platform based on a reliably working sensor and prepare it for real-world deployment.

Embedded

- Get the prototype running reliably on battery
- Implement deep sleep functionality in the prototype firmware
- Ensure Grafana is fully working with live data
- Update and clean up prototype documentation

GitLab & CI Setup

- Set up CI pipeline
- Reference or migrate relevant earlier work

Data Pipeline (MQTT)

Architecture:

Sensor → MQTT Broker → Storage → Consumers

Tasks:

- Deploy MQTT broker
- Build Service
- Document full data flow with a diagram

Custom PCB Design

- Design integrated PCB
- Review with team
- Order prototype

Documentation

- Update prototype documentation continuously
- Ensure all configurations, schemas, and system diagrams are stored in the repository
- Prepare documentation to evolve into DIY kit guides in later sprints

Sprint 2 Deliverables

- GitLab repository with CI pipeline and documented workflow
- MQTT broker running with defined JSON schema
- Working Grafana dashboard with live data
- Prototype running reliably on battery with deep sleep implemented
- Updated and complete prototype documentation
- PCB prototype ordered

Short Summary

Sprint 2: Build and stabilize the system

- Infrastructure (GitLab, MQTT, database, Grafana)
- Reliable battery-powered prototype with deep sleep
- Documentation and coordination
- Preparation for scaling into a DIY kit in Sprint 3–4